Querying enzyme kinetics experiments by substrate and… Geteilt von Jara1991

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Melden

I have some enzyme kinetics experiments stored in the Experiments database. Can you help me find a few things?

First, I'm looking for all experiments where Glucose was used as a substrate at concentrations above 100 mM.

Then I'd like to understand how Pyruvate appears across my experiments — sometimes it's a substrate, sometimes a product. Can you pull both and compare?

Also, which of my experiments were run at 37°C?

And finally, I'm curious about NADH. It shows up in different roles across my experiments. Can you show me everywhere it appears and what role it had?

I'll help you explore your experiments database! Let me start by understanding the schema and relationships, then run all your queries.

Hat 4 Tools verwendet >

Good, now I have the schema. Let me run all your queries in parallel.

Hat 4 Tools verwendet >

Now let me get the parent experiments for context. Let me fetch the experiments linked to these molecules.

Hat 2 Tools verwendet >

Now let me get the experiments for the remaining NADH entries and the 37°C experiments.

Hat 2 Tools verwendet >

Here's the full picture of your experiments database:

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1. Glucose as substrate (>100 mM) — 7 experiments

Experiment	Enzyme	[Glucose]	Temp	pH
Hexokinase Glucose Phosphorylation – Medium Substrate	Hexokinase (EC 2.7.1.1)	150 mM	25°C	7.4
Hexokinase Glucose Phosphorylation – High Substrate	Hexokinase (EC 2.7.1.1)	500 mM	25°C	7.4
Glucose Oxidase – Medium Glucose	Glucose Oxidase (EC 1.1.3.4)	120 mM	30°C	7.5
Glucose Oxidase – High Glucose	Glucose Oxidase (EC 1.1.3.4)	300 mM	30°C	7.5
Hexokinase – G6P Product Inhibition	Hexokinase (EC 2.7.1.1)	150 mM	25°C	7.4
Glucose Isomerase – Medium Glucose	Glucose Isomerase (EC 5.3.1.5)	200 mM	37°C	7.0
Glucose Isomerase – Saturating Glucose	Glucose Isomerase (EC 5.3.1.5)	1000 mM	37°C	7.0

The range spans 120–1000 mM across three enzyme systems: Hexokinase phosphorylation, Glucose Oxidase oxidation, and Glucose Isomerase isomerization.

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2. Pyruvate — dual role comparison

As product (2 experiments): Generated by Pyruvate Kinase (EC 2.7.1.40) from PEP at 37°C/pH 7.0 — both at 0 mM initial (product accumulation studies).

Experiment	[PEP] substrate	Temp/pH
Pyruvate Kinase – Low PEP	75 mM	37°C / 7.0
Pyruvate Kinase – High PEP	200 mM	37°C / 7.0

As substrate (4 experiments):

Experiment	Enzyme	[Pyruvate]	Temp/pH
Gluconeogenesis – Pyruvate to Glucose (Low)	Pyruvate Carboxylase (EC 6.4.1.1)	100 mM	37°C / 6.8
Gluconeogenesis – Pyruvate to Glucose (High)	Pyruvate Carboxylase (EC 6.4.1.1)	250 mM	37°C / 6.8
LDH – Pyruvate to Lactate (Low)	Lactate Dehydrogenase (EC 1.1.1.27)	50 mM	25°C / 8.0
LDH – Pyruvate to Lactate (High)	Lactate Dehydrogenase (EC 1.1.1.27)	200 mM	25°C / 8.0

This nicely captures the metabolic duality: Pyruvate is the *product* of glycolysis (Pyruvate Kinase) but the *substrate* for both gluconeogenesis (Pyruvate Carboxylase) and lactate fermentation (LDH).

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3. Experiments at 37°C — 10 experiments

All your physiological-temperature experiments:

Experiment	Enzyme	pH
Pyruvate Kinase – Low PEP	Pyruvate Kinase	7.0
Pyruvate Kinase – High PEP	Pyruvate Kinase	7.0
Gluconeogenesis – Low Pyruvate	Pyruvate Carboxylase	6.8
Gluconeogenesis – High Pyruvate	Pyruvate Carboxylase	6.8
ADH Ethanol Oxidation – Low Ethanol	Alcohol Dehydrogenase	7.2
ADH Ethanol Oxidation – High Ethanol	Alcohol Dehydrogenase	7.2
Glucose Isomerase – Medium Glucose	Glucose Isomerase	7.0
Glucose Isomerase – Saturating Glucose	Glucose Isomerase	7.0

ATPase Hydrolysis – Medium ATP	ATPase	7.4
ATPase Hydrolysis – High ATP	ATPase	7.4

pH ranges from 6.8 to 7.4 — all near-physiological. These cover 5 different enzyme systems.

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4. NADH — all appearances across 9 molecule entries

As cofactor (5 entries, 3 enzyme systems):

Experiment	Enzyme	[NADH]
Glucose Oxidase – Low Glucose	Glucose Oxidase	5 mM
Glucose Oxidase – Medium Glucose	Glucose Oxidase	5 mM
Glucose Oxidase – High Glucose	Glucose Oxidase	5 mM
Gluconeogenesis – Low Pyruvate	Pyruvate Carboxylase	20 mM
Gluconeogenesis – High Pyruvate	Pyruvate Carboxylase	20 mM

As substrate (2 entries, LDH reactions):

Experiment	[NADH]	Co-substrate
LDH – Pyruvate to Lactate (Low)	50 mM	Pyruvate 50 mM
LDH – Pyruvate to Lactate (High)	200 mM	Pyruvate 200 mM

As product (2 entries, ADH reactions):

Experiment	[NADH] initial	Note
ADH Ethanol Oxidation – Low Ethanol	0 mM	Generated from NAD+
ADH Ethanol Oxidation – High Ethanol	0 mM	Generated from NAD+

NADH plays three distinct roles across your experiments — cofactor in oxidation/gluconeogenesis, co-substrate consumed by LDH, and product generated by ADH. The concentration range when supplied spans 5–200 mM, with the LDH experiments using notably higher concentrations than the cofactor applications.

Want me to dig deeper into any of these, or perhaps create a summary spreadsheet?



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